

3/9/23

Biology

1. ATP and ~~na~~ NADPH together called Assimilatory Power connects light reaction with dark reaction.
2. Aerobic respiration occurs in the mitochondria in eukaryotic cells.
3. The lymph is devoid of RBCs (Red Blood cells)
4. The plants gets rid of ~~ex~~ excess water through the process called transpiration.
5. ~~Malp~~ Malpighian body is formed by Glomerulus and Bowman's capsule
6. Chlorophyll content of leave, ~~humid~~ age of plant, etc does not affect contribute to rate of transpiration.
7. i) Plants do not breathe, they only respire through leaves.
Whereas
Animals breathe air for cellular respiration.

(ii) Plants obtain O_2 directly from the air through stomata.

Whereas
Animals inhale O_2 from nose or gills into their respective respiratory tract organ.

8.	Light reaction	Dark reaction
	* Light dependent.	* Light interdep independent.
	* occurs in grana	* occurs in stroma.

9(a) Pepsin and Trypsin acts on protein.

(b) In absence of oxygen pyruvate undergo reduction and produce lactate.

11(a) Alveoli is covered with blood capillaries so that the exchange of gases takes place between the membrane of alveoli and the surrounding capillaries.

(b) The wall of trachea is supported

with ~~at~~ cartilaginous rings as it supports the trachea and prevent it from collapsing.

12(a) During winter, the surrounding temperature is very low. So there is no sweating and there is no loss of water in any other forms. Due to this reason we pass more urine in winter season.

(b) Lungs, Liver, skin and intestine are the ~~ex~~ accessory excretory organs in humans.

(c) Major organs of excretion in ~~Arthro~~ Arthropods are green gland and malpighian tubules.

13. (i) '4' is left atrium

'5' is opening of inferior vena cava (Right atrium)

(ii) '3' is superior vena cava which carries deoxygenated ^{blood} from upper parts of body.

'6' is inferior vena cava which carries oxygen-poor blood from lower part of body.

(iii) Pulmonary artery carries blood away from heart.

(iv) The blood circulation in human is called double circulation because the blood passes through the heart twice in one complete cycle of the body.

* The oxygenated blood is separate from deoxygenated blood.

* Pulmonary circulation and systematic circulation takes place in one cycle.